Problem 2.1 (2 pts) L8 3D Trim

Estimate the maximum airspeed for the jet aircraft if it is flying a constant altitude of 4.5 km.

mass, m = 12000 kg

wing reference area, S = 29 m^2

CD = CD0 + K*CL*CL
CD0 = 0.04 + 0.02*tanh[5.0*(M-0.95)]
K = 0.7 + 0.2*tanh[5.0*(M-0.95)]

Engine: (M = Mach number, h = altitude [km], T = max thrust [kN])
T = 78.7 + 13.3*M - 6.3*h + 0.13*h*h + 7.3*M*M – h*M ← convert to N

Standard Atmosphere at 4.5 km:

density, rho = 0.78 kg/m^3
speed of sound: sos = 322 m/s, M = V/sos

maximum airspeed = 334.95 [m/s]

Hint(s):

Slide 7 Boxed Eqn

↳ Plot as function of V (y-axis) until V becomes constant

• Can also plot T + M functions + find where they intersect

↑ point of max speed

Equation 1:

$$T = 78.7 + 13.3M - 6.3h + 0.13h^2 + 7.3M^2 - hM \quad \text{where } M = \frac{V}{322},\ h = 4.5\,\text{km}$$

↳ will be in [kN] ∴ $T_1 = T(1000)$ to ensure output is in [N]

Equation 2:

$$T_2 = \frac{1}{2}\rho V^2 S C_{D0} + \frac{2KW^2}{\rho V^2 S} \quad \text{where } \rho = 0.78\ \text{kg}/\text{m}^3,\ S = 29\,\text{m}^2$$

$C_{D0} = 0.04 + 0.02\tanh\left(5\left(M - 0.95\right)\right)$

$K = 0.7 + 0.2\tanh\left(5\left(M - 0.95\right)\right)$

$W = (12{,}000\,\text{kg})(9.8\,\text{m/s}^2)$

Max speed found when thrust eqns set equal to each other:

$$T_1 = T_2$$

Problem 2.2 (2 pts)

The first flight of Blue Origin's New Glenn rocket took place on 16 JAN 2025. The first stage of the rocket is equipped with seven (7) BE-4 motors. Each motor has a specific impulse of 340 seconds and produces 2,400 kN of thrust at sea level. The second stage is powered by two (2) BE-3U motors that each provide 770 kN of thrust with a specific impulse of 445 seconds. Simulate the rocket's trajectory for 800 seconds after the initial time using a Runge-Kutta method and a step size of one (1) second. Plot the simulated vehicle airspeed [km/s] on the y-axis against time [s] on the x-axis. Compare your simulated airspeed by also plotting the reported airspeed trajectory. Be sure to label each axis and include a legend to mark each curve. Submit your plot using a standard image format (PNG, JPG, BMP).

$$\frac{dV}{dt}=\frac{T}{m}-g_0\left(\frac{R_E}{R_E+h}\right)^2 \sin\gamma$$

From Non-Rotating Spherical Earth Slide (L9)

$$\tau\frac{d\gamma}{dt}+\gamma=0$$

assume 90° → 0°

$$\frac{dh}{dt}=V\sin\gamma$$

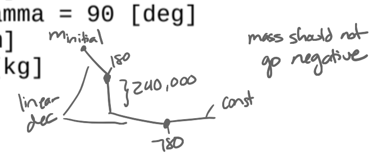
initial angle

$$\frac{dm}{dt}=-\frac{T}{g_0 I_{sp}}$$

RE = earth radius = 6378 km
g0 = sea-level gravity = 9.8 m/s^2
tau = guidance time constant = 120.0 sec

A. initial conditions:

time, t = 45 [sec]
airspeed, V = 45 [m/s]
flight path angle, gamma = 90 [deg]
altitude, h = 1600 [m]
mass, m = 1,200,000 [kg]



B. flight events:

(1) Main Engine Cut-Off (time = 180 sec)
  a. shut down first stage motors
  b. drop first stage structure, m = 240,000 [kg]
  c. start second stage motors
(2) Second-Stage Cut-Off (time = 780 sec)
  a. shut down second stage motors

C. comparison data file (ng1.csv) ← given on Canvas

column #1 = time [sec]
column #2 = aispeeed [m/s]

Hint(s):
- L9 SimX15
- Use simx15.m code for guidance
- Any Runge-Kutta order allowed
- step size: 1 sec

Don't need standard atmosphere function; can start from Runge-Kutta eqns

*ensure that you drop the mass correctly

Problem 2.3 (2 pts) L8 Slide 9

An airplane is beginning its final approach to an airport at an airspeed of 33 m/s. Air Traffic Control has requested the pilot to perform a “standard rate turn”; that is, to turn at a rate that would result in a 360 degree change in flight path heading (chi) in two minutes. What bank angle (mu) is required for this airplane to fly a standard rate turn at constant altitude?

a. 2.5 deg

b. 10 deg

c. 68 deg

d. 84 deg

$V = 33\,m/s$ (airspeed)

$$\frac{d\chi}{dt} = \frac{360\ deg}{2\ min} \times \frac{1\ min}{60\ s} \times \frac{2\pi\ rad}{360\ deg} \Longrightarrow \frac{d\chi}{dt} = \frac{\pi}{60}\ \frac{rad}{s}$$

$g = 9.8\ m/s^2$

$$\tan\mu = \frac{V}{g}\frac{d\chi}{dt}$$

$$\mu = \tan^{-1}\left(\frac{V}{g}\frac{d\chi}{dt}\right) \quad [rad]$$

$\uparrow$

multiply by $\frac{180}{\pi}$ to get answer in [deg]

Problem 2.4 (2 pts) L10 Slide 9 (plug & chug into b eqns)

An aircraft simulation uses quaternions to represent the relationship between the local and body-fixed reference frames. Find the four quaternion values that are needed to set the initial aircraft state so that the equivalent Euler angles are: roll attitude (phi) = -45 deg, pitch attitude (theta) = +10 deg, and heading (psi) = 90 deg.

a. (b0,bx,by,bz) = (-0.123,-0.123,-0.696,0.696)

b. (b0,bx,by,bz) = (0.674,0.213,0.327,0.627)

c. (b0,bx,by,bz) = (-0.528,-0.640,0.558,0.035)

d. (b0,bx,by,bz) = (0.627,-0.327,-0.213,0.674)

$\phi = -45$ deg

$\theta = 10$ deg

$\psi = 90$ deg

plug in here $\longrightarrow$

$$b_0 = \cos(\phi/2)\cos(\theta/2)\cos(\psi/2) + \sin(\phi/2)\sin(\theta/2)\sin(\psi/2)$$

$$b_x = \sin(\phi/2)\cos(\theta/2)\cos(\psi/2) - \cos(\phi/2)\sin(\theta/2)\sin(\psi/2)$$

$$b_y = \cos(\phi/2)\sin(\theta/2)\cos(\psi/2) + \sin(\phi/2)\cos(\theta/2)\sin(\psi/2)$$

$$b_z = \cos(\phi/2)\cos(\theta/2)\sin(\psi/2) - \sin(\phi/2)\sin(\theta/2)\cos(\psi/2)$$

Problem 2.5 (2 pts) L11 Same procedures as Slide 4

Which set of differential equations can be used to simulate pure rolling motion (i.e. q = r = 0)?

a. $\frac{db_0}{dt} = -\frac{1}{2} b_x p$ and $\frac{db_x}{dt} = \frac{1}{2} b_0 p$ and $\cancel{\frac{db_0}{dt}} = \frac{db_x}{dt} = 0$

b. $\frac{db_0}{dt} = -\frac{1}{2} b_z r$ and $\frac{db_z}{dt} = \frac{1}{2} b_0 r$ and $\cancel{\frac{db_x}{dt}} = \frac{db_y}{dt} = 0$

c. $\frac{db_0}{dt} = -\cancel{\frac{1}{2} b_0 p}$ and $\frac{db_x}{dt} = \frac{1}{2} b_x p$ and $\frac{db_y}{dt} = \frac{db_z}{dt} = 0$

d. $\frac{db_0}{dt} = -\frac{1}{2} b_x p$ and $\frac{db_x}{dt} = \frac{1}{2} b_0 p$ and $\frac{db_y}{dt} = \frac{db_z}{dt} = 0$

$$\begin{bmatrix} \dot{b}_0 \\ \dot{b}_x \\ \dot{b}_y \\ \dot{b}_z \end{bmatrix} = \frac{1}{2} \begin{bmatrix} -b_x & -b_y & -b_z \\ b_0 & -b_z & b_y \\ b_z & b_0 & -b_x \\ -b_y & b_x & b_0 \end{bmatrix} \begin{bmatrix} p \\ q \\ r \end{bmatrix}$$

Pure rolling motion: $q = r = b_y = b_z = 0$

$$\begin{bmatrix} \dot{b}_0 \\ \dot{b}_x \\ \dot{b}_y \\ \dot{b}_z \end{bmatrix} = \frac{1}{2} \begin{bmatrix} -b_x & 0 & 0 \\ b_0 & 0 & 0 \\ 0 & b_0 & -b_x \\ 0 & b_x & b_0 \end{bmatrix} \begin{bmatrix} p \\ 0 \\ 0 \end{bmatrix}$$